

TASK 5 — EXCEL DATA ANALYSIS

VEDA Technology Internship | Level 1

This report documents the Excel-based analysis performed on the Sample Superstore dataset. The workbook contains a protected raw-data layer, a working/formula layer, a KPI summary, and five business-oriented PivotTable-style analyses.

1. Dataset Overview

Item	Value
Dataset	Sample Superstore
Records	10,194
Unique Orders	5,111
Date Range	03-Jan-2023 to 30-Dec-2026
Total Sales	2,326,534.35
Total Profit	292,296.81
Total Quantity	38,654

2. KPI Analysis

Total sales were 2,326,534.35, while total profit was 292,296.81. The workbook records 5,111 unique orders and 38,654 units. The calculated average order value is 455.20, with an overall profit margin of 12.56%.

3. Business Questions Answered

- Which category generates the most sales?
- Which region generates the most sales?
- How do sales change month by month?
- Which sub-categories generate the strongest sales and profit?
- Which customer segment contributes the most sales?

4. Category Analysis

Category	Sales	Profit	Quantity	Margin
Technology	839,893.28	146,543.38	7,017	17.45%
Furniture	754,747.76	19,730.00	8,369	2.61%
Office Supplies	731,893.31	126,023.44	23,268	17.22%

Finding: Technology is the highest-sales category.

5. Regional Analysis

Region	Sales	Profit	Quantity	Margin
West	739,813.61	110,798.82	12,466	14.98%
East	691,828.17	94,883.26	11,159	13.71%
Central	503,170.67	39,865.31	8,820	7.92%
South	391,721.91	46,749.43	6,209	11.93%

Finding: West is the highest-sales region.

6. Sub-Category and Segment Analysis

The strongest sub-category by sales is Chairs, while Copiers has the highest total profit. The workbook also includes a segment-level analysis to compare Consumer, Corporate, and Home Office performance.

7. Monthly Trend Analysis

The Pivot_Month sheet summarizes sales, profit, and quantity by month and includes a line chart for visual trend analysis. This allows management to identify periods of stronger demand and investigate changes in profitability over time.

8. Excel Techniques Used

- Structured Excel tables and filters
- SUMIFS for conditional sales aggregation
- COUNTIFS for conditional record/order analysis
- AVERAGEIFS for conditional average analysis
- PivotTable-style aggregation by category, region, month, sub-category, and segment
- KPI calculations and business findings
- Bar and line charts for visual analysis

9. Business Summary

The analysis indicates total sales of 2,326,534.35 and total profit of 292,296.81. Technology leads category sales and West leads regional sales. At product-group level, Chairs is the highest-sales sub-category and Copiers is the most profitable. The combination of formula-based calculations and PivotTable analysis provides a repeatable approach for identifying sales concentration, profitability, and time-based performance patterns.